pioneer in urban air mobility (UAM). Beginning in 2017, the Dubai Roads and Transport Authority (RTA), in collaboration with companies such as EHang and Volocopter, conducted full-scale demonstration flights of drone taxis intended for short-distance intra-city travel. These early trials showcased the feasibility of electric vertical take-off and landing (eVTOL) aircraft operating along designated aerial corridors to transport passengers between strategic urban hubs without conventional runways.

Dubai's approach to drone taxi deployment reflects a policy-first and infrastructure-driven strategy. The city-initiated planning for an integrated network of vertiports – compact aerial mobility stations located at transportation nodes, rooftops, and business districts. These vertiports are designed to provide passenger handling, safety procedures, rapid charging facilities, and digital interfaces linking aerial vehicles to the city's broader public transportation grid. The goal is the development of a multi-layered mobility ecosystem, where ground and air transportation operate in seamless coordination.

The early demonstrations also highlighted Dubai's commitment to regulatory innovation. The city worked proactively with aviation authorities to draft operational safety requirements, passenger protection standards, autonomous traffic management rules, and airspace zoning guidelines for low-altitude flight. Although full commercial rollout is still in development, Dubai's demonstration flights served to build public awareness and global confidence in the practicality and safety of passenger drones.

Figure 9.7 illustrates the early adoption of drone taxi technology in Dubai. A multi-rotor passenger eVTOL aircraft is shown approaching a rooftop vertiport, set against the modern Dubai skyline. These demonstration flights showcased Dubai's pioneering efforts in urban air mobility (UAM), emphasizing infrastructure planning, regulatory readiness,